Hriday Sarkar

Embedded Systems Engineer | IoT Researcher | Robotics Developer

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with hands-on experience in embedded systems, IoT, AI-integrated automation, robotics, and business development. Skilled in developing ESP32/ESP8266-based smart systems, cloud-connected IoT platforms, autonomous robots, and energy-efficient electronic solutions. Experienced in leading technical and business operations through roles in e-commerce and technology-driven projects, with strong expertise in system analysis, automation, hardware-software integration, and product innovation. Passionate about building sustainable, AI-supervised technologies and transforming innovative ideas into practical real-world solutions through research, development, and entrepreneurship.

WORK EXPERIENCE

echithi — **Dhaka, Bangladesh**
Business Development Team Lead — *2023 – Present*

- Lead corporate growth strategies for a robotics-centric tech company, managing B2B partnerships, client outreach, and product roadmaps to expand market reach.
- Spearheaded the Phoenix Project, a national-level embedded systems, IoT, and AI initiative designed to accelerate industrial automation and institutional adoption.
- Coordinated cross-functional teams of [X] hardware engineers, software developers, and sales personnel to successfully design and deliver integrated robotic solutions.
- Founded, built, and launched the Born To Code (BTCs) educational program, training over [X] school students in hardware-software integration and foundational C/C++ programming.

echithi — **Dhaka, Bangladesh**
System Analysis and Data Entry Executive — *2022 – 2023*

- Conducted hardware-software system analysis and managed structured performance reporting pipelines for active robotics product lines.
- Structured and maintained technical product documentation, functional specifications, and deployment schematics for cloud-connected IoT systems.
- Executed early-stage prototyping assessments, functional QA testing, and rigorous hardware debugging cycles for custom ESP32/Arduino embedded devices.

EDUCATION

BSc in Computer Science and Engineering — *2019 – 2023*
University of Asia Pacific, Bangladesh — Dhaka, Bangladesh

Higher Secondary Certificate (HSC) — *2017*
Ramganj Model Degree College, Ramganj

Secondary School Certificate (SSC) — *2015*
Ramganj High School, Ramganj

UNDERGRADUATE THESIS

Garbage Sorting System with Image Processing Using CNN

Supervised by Assistant Professor A. S. Zaforullah Momtaz, University of Asia Pacific

Designed and implemented a CNN-based intelligent garbage classification system capable of identifying waste categories using image processing techniques for automated waste management applications.

RESEARCH INTERESTS

- Robotics and Autonomous Systems
- Internet of Things (IoT) Architectural Design
- Embedded Firmware Development
- Machine Learning, TinyML and Edge Computer Vision
- Industrial Automation & Sensor Networks
- Renewable Energy Harvesting Systems
- AI-assisted Edge Intelligent Devices

TECHNICAL SKILLS

Microcontrollers & Hardware: ESP32, ESP8266, Arduino, NRF24L01, RFID/NFC, RTC modules, DHT22, NeoPixel (WS2812B), SD Card modules, relay modules, IR sensors, electrochemical gas sensors

Communication Protocols: Wi-Fi (AP + STA Mode), HTTP/HTTPS, MQTT, SPI, I2C, UART, NRF24L01 RF, Over-The-Air (OTA) firmware updates

Programming Languages: C/C++ (Arduino framework), Python, JavaScript, LaTeX

IoT & Cloud: ESP32 EEPROM credential management, Network Time Protocol (NTP) sync, server-push automation, REST API integration, web-based device configuration interfaces (Web UIs)

AI & Software: Edge-ML integration for embedded systems, biometric/fingerprint authentication frameworks

Robotics: Multi-domain mechanical/firmware robot design (gas detection, firefighting, surveillance, gesture control), warehouse navigation kinetics, ROS-compatible architecture

Agriculture Technology: Hydroponic fodder automation, ESP32-based multi-tier vertical farming, automated closed-loop pH, humidity, and custom irrigation management

Tools & Platforms: Git, GitHub, VS Code, Arduino IDE, Microsoft Office Suite

KEY PROJECTS

ESP32 EEPROM-Based Wi-Fi Credential Management System

- Engineered a robust credential fallback system with auto-connect routine, persistent access-point (AP) configuration dashboard, NeoPixel status arrays, and automated RSSI/LRU network prioritized connection fallback.

RM22 IoT Multi-Domain Robot Platform

- Architected a modular mobile robot payload system configured dynamically for real-time gas leakage maps, autonomous fire suppression maneuvers, and secure IP-camera video streaming over local access points.

ESP32 Integrated Biometric Signature System

- Built an enterprise full-stack IoT attendance logger blending high-speed RFID reads, local offline SD card logging arrays, precise RTC timekeeping, NTP server validation, and background HTTP web dashboard uploads.

inCook Portable Smart Induction/Infrared Cooker

- Co-developed a consumer hardware MVP featuring intelligent firmware power balancing, Bluetooth/Mobile App pairing profiles, and standalone off-grid Li-Ion/solar charger integration.

Automated Vertical Hydroponic Fodder System

- Developed a 4-layer precision vertical farming controller utilizing low-power ESP32 logic, real-time chemical pH tracking loops, industrial humidity arrays, and strict cron-scheduled solenoid valve irrigation.

SD Card Credentials Dashboard & Dual-OTA Architecture

- Constructed a secure configuration subsystem pulling cryptographic Wi-Fi profile arrays from local SD cards alongside failsafe dual-partition Over-The-Air updates using AP fallback options and routine device-to-cloud server push actions.

ONGOING RESEARCH & PUBLICATIONS

- **Autonomous IoT Robot Based on RM22 Platform: A Multi-Layer Wireless Control System for Smart Application** – Multi-domain IoT robotics architecture; manuscript targeted for peer-reviewed submission to IEEE, MDPI, or Springer publications.
- **Towards Smarter Low-Cost Robotics: Design and Evaluation of the CM14 Capsule Mover** – Engineering design analysis of an ultra-low-cost, internet-controlled industrial mobile platform.

WORKSHOPS, AWARDS & CERTIFICATIONS

- **Lead Technical Trainer**, CM14 Series – Advanced IoT Powered Smart Robot Workshops (2026)
- **Lead Technical Trainer**, RM22 Series – Advanced IoT Based Smart Robot Workshops (Delivered 5+ Sessions) (2025)
- **Assistant Technical Workshop Trainer**, RM22 Embedded Systems Core Curriculum (2024)
- **Technical Co-Trainer**, Arduino Uno R3 – Mastering Microcontroller Architectures & Hardware Control (2024)
- **Technical Co-Trainer**, BM12 – Intelligent Closed-Loop Obstacle Avoidance Robotics Workshop (2024)
- **Honorable Mention**, JRC Memorial Math Fest Intra-University Mathematics Olympiad (2023)

LANGUAGES

Bengali (Native) | English (Professional Working Proficiency) | Hindi (Basic working proficiency)